DO NOW — RECALL

Lesson 062 · Thinking Procedurally: Ordered Subtasks

Name: _____ Class: _____ Date: _____

1. What is a precondition?

2. What is decomposition?

3. What is a dependency in a process?

4. An empty stack executes push(3), push(5), x=pop(), push(8), then push(x+peek()). peek() returns the top value without removing it. Trace the final stack from bottom to top.

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

Rule that must hold before an operation.

2. Model answer

Breaking a problem into manageable subtasks.

3. Model answer

A stage requiring another stage's completed result.

4. Model answer

3, 8, 13. x is 5; peek() is 8, so the final push adds 13.

By the end, you can…

• Identify the separate components of a problem: validation, stock check, updating records and confirming a result.

• Determine necessary order from dependencies. Do not send success or deduct stock before validation.

• Decompose large operations into subprocedures, each with clear responsibility; update-related steps may need a single transaction to maintain consistency.

Vocabulary — know these terms

procedural thinking — Planning ordered steps.

subprocedure — Smaller named piece of a solution.

dependency — A stage depends on another.

atomicity — All required updates succeed or none do.

NEW LEARNING

Read, annotate and explain

Identify

Identify the separate components of a problem: validation, stock check, updating records and confirming a result.

Model

Determine necessary order from dependencies. Do not send success or deduct stock before validation.

Evaluate

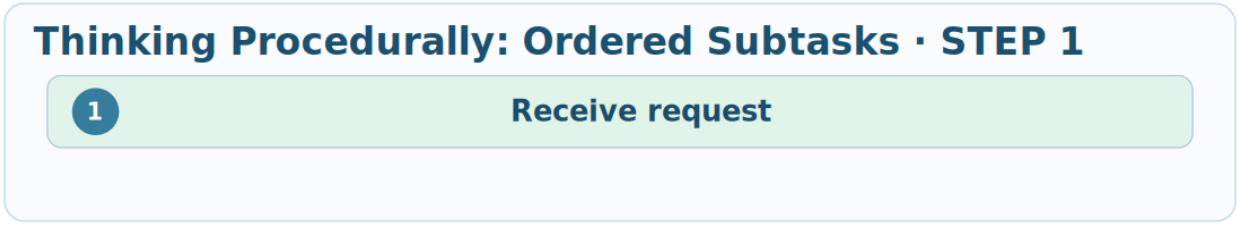
Decompose large operations into subprocedures, each with clear responsibility; update-related steps may need a single transaction to maintain consistency.

Explain one step in your own words, using the correct technical terms.

New Learning · original lesson diagrams

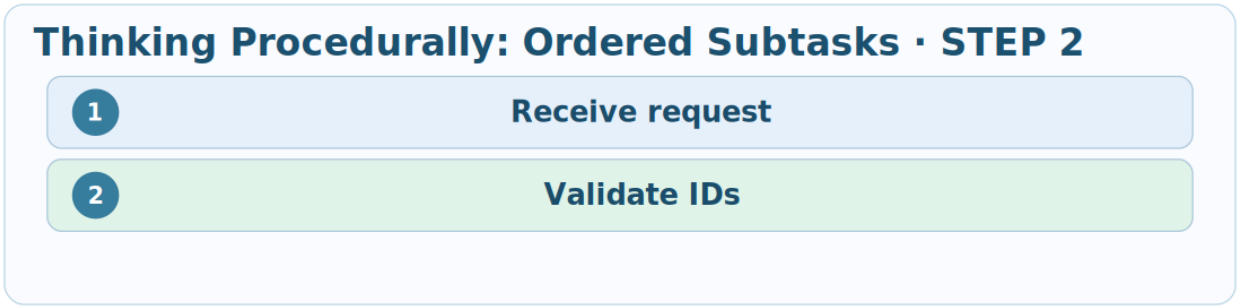
Diagrams appear after the related explanation. Annotate the original source visual.

New Learning



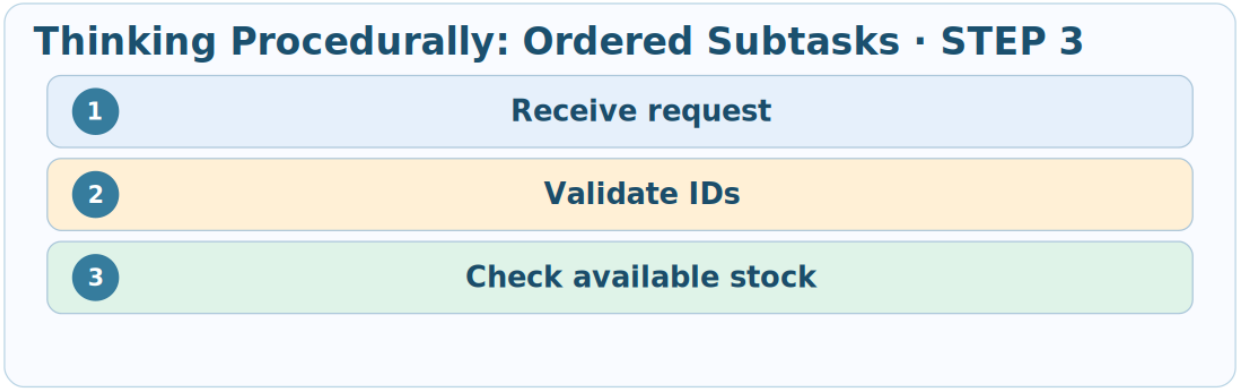
Thinking Procedurally: Ordered Subtasks · STEP 1: Receive request

New Learning



Thinking Procedurally: Ordered Subtasks · STEP 2: Receive request; Validate IDs

New Learning



Thinking Procedurally: Ordered Subtasks · STEP 3: Receive request; Validate IDs; Check available stock

New Learning · original lesson diagrams

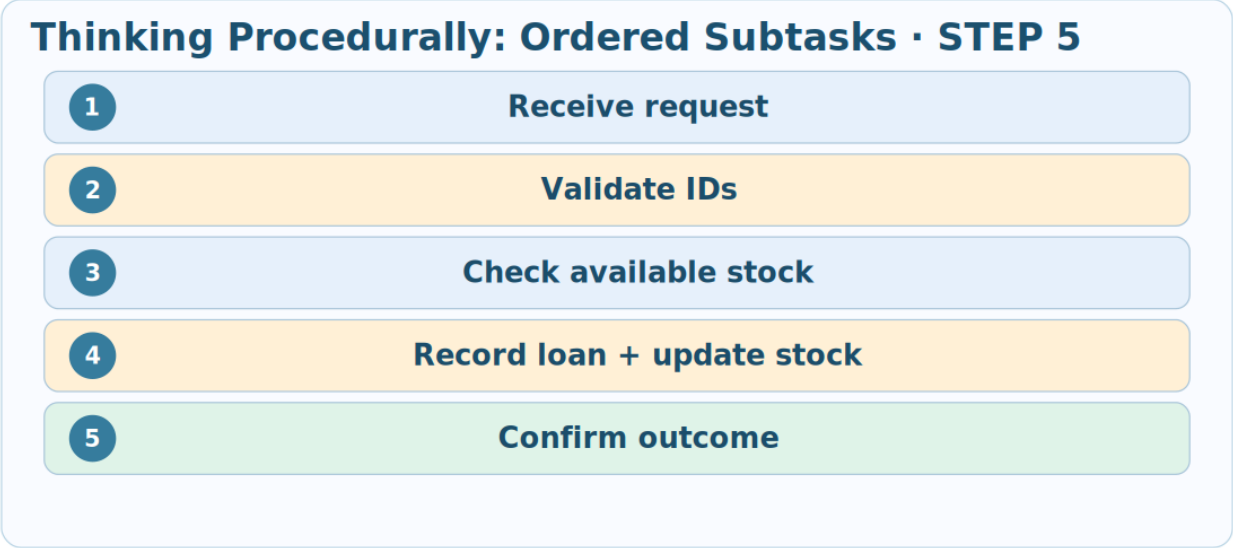
Diagrams appear after the related explanation. Annotate the original source visual.

New Learning



Thinking Procedurally: Ordered Subtasks · STEP 4: Receive request; Validate IDs; Check available stock; Record loan + update stock

New Learning



Thinking Procedurally: Ordered Subtasks · STEP 5: Receive request; Validate IDs; Check available stock; Record loan + update stock; Confirm outcome

I DO — WORKED EXAMPLE

Follow the worked method

Put steps in order to process an exam result securely: check identity, find result, return data.

Explain the decision

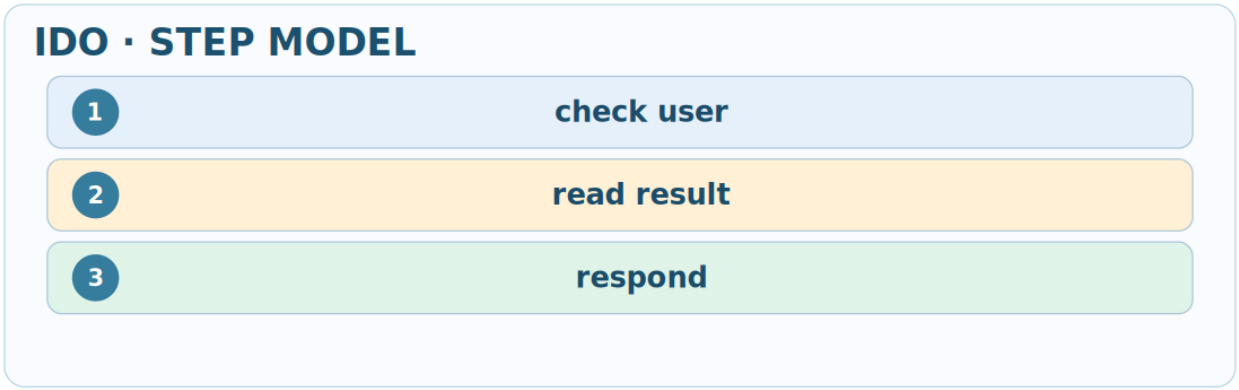
If lookup occurs before checking authorisation, data can be exposed through insecure responses.

I Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

Put steps in order to process an exam result securely: check identity, find result, return data.

If lookup occurs before checking authorisation, data can be exposed through insecure responses.



IDO · STEP MODEL: check user; read result; respond

WE DO — GUIDED PRACTICE

Try → Discuss → Improve

TRY: Attempt the guided questions in BLACK PEN. DISCUSS: Compare approaches with a partner, explain your reasoning and agree one improvement. IMPROVE: After the Exit Ticket, use the source CHECK model in GREEN PEN to correct and explain errors.

Guided 1 — TRY

A checkout program has steps deduct stock, check payment, pack goods. Choose order and justify.
Source code / input:
Validate order → authorise payment → atomically confirm order and stock → pack or dispatch.

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations or a justified explanation.

Question 1

Order tasks: issue library loan, check stock, validate pupil, generate receipt. Explain first and last. [6]

Question 2

Write three helper procedure names for the library process, each with one responsibility. [6]

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations or a justified explanation.

Question 3

A system sends confirmation before saving a booking, then crashes. Explain fault and propose safer steps. [6]

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking.

A confirmation should be sent before a transaction succeeds, to make an application feel faster.

Circle: Confident / Getting there / Need help

Explain what you can now do independently.

GREEN PEN — Correct and explain

Choose one error. Write the correct method and explain precisely why your first response was wrong.

WE DO — CHECK AND IMPROVE

Source-grounded model answers

CHECK only after completing the guided practice and Exit Ticket. Use GREEN PEN to improve your method and explain any error.

Guided 1 — CHECK

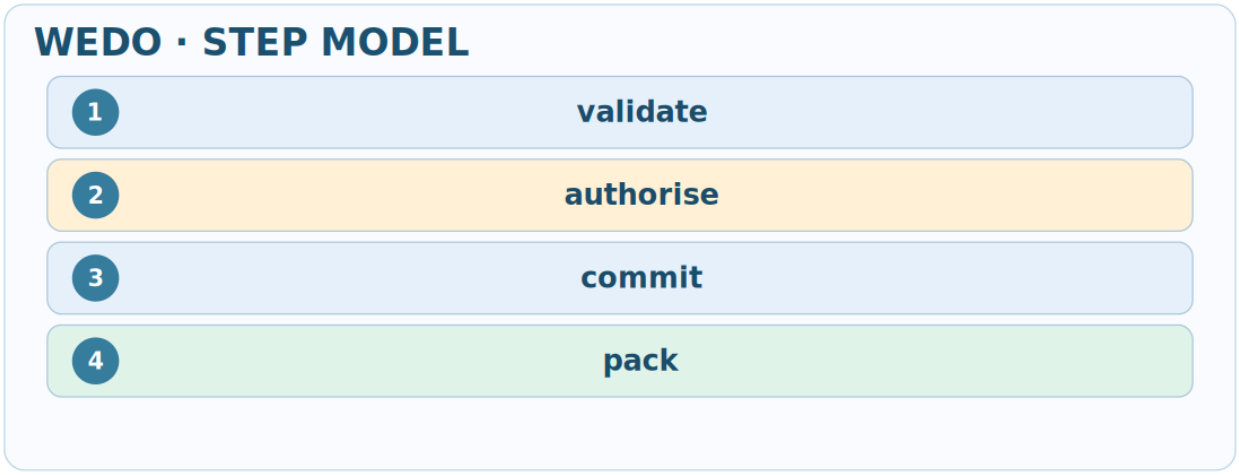
Explain the decision
Payment and stock operations need failure handling; no dispatch before a confirmed order.

We Do source CHECK · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

A checkout program has steps deduct stock, check payment, pack goods. Choose order and justify.

Payment and stock operations need failure handling; no dispatch before a confirmed order.



WEDO · STEP MODEL: validate; authorise; commit; pack